

What Really Matters in First-Shot Anomalous Sound Detection? An Empirical Study of Encoders, Adaptation, and Anomaly Scoring

Huan Yu, Yanjin Li, Zhongxin Bai, Gongping Huang*

Wuhan University, Wuhan, China,

yuhappy2007@outlook.com, liyanjin@whu.edu.cn, zhongxinbai@whu.edu.cn, gongpinghuang@whu.edu.cn

Abstract—We present an empirical study of the factors that affect performance in first-shot unsupervised anomalous sound detection (ASD), based on our system developed for DCASE 2026 Task 2. We organize the study into two stages consisting of representation learning and anomaly scoring. For representation learning, we examine pretrained audio encoders, layer selection, encoder ensembling, and LoRA-based adaptation. For anomaly scoring, we investigate whitening-based normalization and several scoring back-ends, including local-density k -nearest neighbors (k NN), relative Mahalanobis distance (RMD), and their combination. Our results show that aggregating selected layers outperforms using only the final layer on the DCASE 2026 Task 2 development set. LoRA-based adaptation primarily improves cross-domain performance, while its in-domain effect varies across encoders. The effect of encoder ensembling depends on model composition, and adding more encoders does not necessarily improve performance. For anomaly scoring, whitening substantially improves performance, with local-density k NN providing the strongest individual back-end among those evaluated. Combining it with RMD further improves in-domain performance, whereas standalone RMD performs close to the chance level. The submitted system achieves 66.65% on the DCASE 2026 Task 2 development set and 65.45% on the DCASE 2026 Task 2 official evaluation set. These results are obtained using only near-field, single-channel audio.

Index Terms—anomalous sound detection, first-shot unsupervised learning, encoder ensemble, LoRA, domain generalization

1. INTRODUCTION

First-shot unsupervised anomalous sound detection (ASD) aims to detect machine anomalies using only normal training data while generalizing to unseen machine types and acoustic environments [1]. The key challenge therefore lies not only in detecting anomalies from limited normal data, but also in learning transferable representations and robust anomaly-scoring strategies that can generalize across domain shifts caused by differences in machine types, operating conditions, and acoustic environments [1]–[4].

Recent studies on first-shot ASD have improved different components of the overall detection pipeline. For representation learning, prior studies explore self-supervised learning, large-scale audio pre-training, and task-specific adaptation to obtain more transferable representations of machine sounds [5]–[8]. Because real anomalous recordings are unavailable during development, proxy anomalies are generated through signal simulation, generative models, or machine metadata, for learning anomaly-sensitive representations and decision rules [9], [10]. At the decision stage, existing studies typically organized normal embeddings into memory banks and further adopt strategies such as density-based scoring, domain-aware normalization, score calibration, and system fusion to improve robustness [2], [4], [8], [11]. Despite this progress, these techniques are often evaluated independently or under different experimental settings, making it difficult to determine which design choices generalize across domains and how their effects interact within a complete ASD system.

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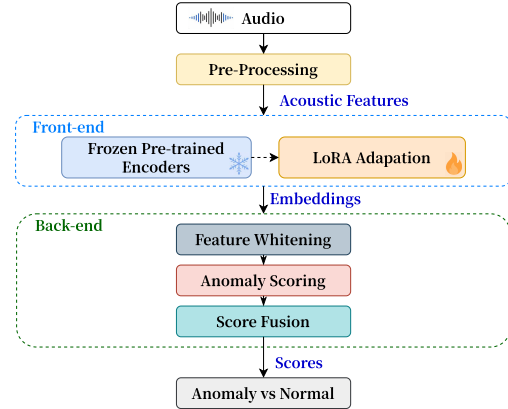


Fig. 1: Framework of the first-shot ASD system.

Based on our DCASE 2026 Challenge Task 2 system [12], we conduct an empirical study of the factors that influence first-shot ASD performance.

Specifically, we analyze representation-learning and anomaly-scoring strategies using in-domain evaluation on the DCASE 2026 Task 2 development set and cross-domain evaluation on the DCASE 2023–2025 Task 2 evaluation sets [1], [13], [14]. The main contributions are summarized as follows:

- We systematically evaluate how encoder choice, layer selection, and parameter-efficient adaptation affect first-shot ASD generalization;
- We identify key factors for designing reliable anomaly scoring strategies under domain shifts;
- We validate the resulting design choices in our DCASE 2026 Task 2 system, which achieves 65.45% on the DCASE 2026 Task 2 official evaluation set using only near-field, single-channel audio.

2. SYSTEM DESCRIPTION

As shown in Fig. 1, we employ a two-stage framework for first-shot ASD. Specifically, given an audio clip, clip-level embeddings are first extracted using pre-trained audio encoders or their LoRA-adapted variants. The resulting embeddings are then converted into anomaly scores using a training-free scoring back-end. The two stages are described in detail below.

2.1. Audio Representation Extraction with Pretrained Encoder

To investigate how the choice of representation model affects first-shot ASD performance, we evaluate six pretrained audio encoders, namely ATST-Frame [15], SSLAM [16], EAT-large [17], BEATs [18], CED [19] and AST [20]. For each audio clip, we extract frame-level embeddings from multiple layers of each encoder and apply temporal mean pooling to selected layers to obtain clip-level representations.

Table 1: Hyperparameters for LoRA adaptation applied to the query and value components of the QKV projections. The “LR”, “EP” and “BS” denote the learning rate, number of training epochs, and batch size, respectively.

Encoder	QKV Types	Rank	LR	EP	BS
SSLAM	Fused	$r = 8$	1×10^{-4}	15	16
EAT-large	Fused	$r = 8$	1×10^{-4}	15	8
BEATs	Separate	$r = 4$	1×10^{-5}	15	16

Instead of relying solely on the final layer or a single optimal layer, we select three high-performing layers for each encoder on the DCASE 2026 Task 2 development set to investigate the impact of layer selection on ASD. When multiple layers are used, an anomaly score is computed separately for each layer and the resulting scores are fused as described in Section 2.3.

2.2. LoRA-based Encoder Adaptation

To adapt the pretrained SSLAM, EAT-large and BEATs to machine sounds while limiting the number of trainable parameters, we apply LoRA adaptation [21]. All pretrained parameters are kept frozen, and only the inserted low-rank matrices are optimized. Specifically, for a pretrained linear projection $\mathbf{W} \in \mathbb{R}^{d \times k}$, the adapted output is

$$\mathbf{h} = \mathbf{W}\mathbf{x} + \mathbf{B}\mathbf{A}\mathbf{x}, \quad (1)$$

where $\mathbf{x} \in \mathbb{R}^{k \times 1}$ denotes the input vector, $\mathbf{A} \in \mathbb{R}^{r \times k}$ and $\mathbf{B} \in \mathbb{R}^{d \times r}$ are trainable low-rank matrices, with $r \ll \min(d, k)$. For SSLAM and EAT-large, LoRA is inserted into the query and value components of the fused QKV projection, with rank $r = 8$. For BEATs, which uses separate attention projections, LoRA is applied to the query and value projections with rank $r = 4$. The remaining adaptation hyperparameters are summarized in Table 1.

We train the adapters using two alternative objectives and compare their performance in Section 3.2. The first is the self-supervised VICReg objective [22], which learns the embedding space from normal clips using invariance, variance, and covariance regularization without requiring labels. The second is a supervised Sub-center ArcFace objective [23], [24], for which a Sub-center ArcFace classification head is applied to the final embedding layer. We use three sub-centers for each class and assign each clip a composite class label defined by its machine type, domain, and attribute. The ArcFace objective is further regularized by a multi-layer mean squared error (MSE) distillation loss [25], which encourages the pooled representations of the adapted encoder to remain close to those of the frozen pretrained encoder at multiple layers. For ArcFace-based adaptation, the overall training objective is

$$\mathcal{L} = \mathcal{L}_{\text{ArcFace}} + \lambda \mathcal{L}_{\text{distill}}, \quad (2)$$

where λ balances the classification and distillation terms.

2.3. Training-free Anomaly Scoring

Before anomaly scoring, each embedding $\mathbf{z}$ is transformed into a whitened feature space [26],

$$\tilde{\mathbf{z}} = \mathbf{P}(\mathbf{z} - \boldsymbol{\mu}), \quad (3)$$

where $\boldsymbol{\mu}$ denotes the mean of the source-domain normal training embeddings, and $\mathbf{P}$ is the whitening transform estimated from the same data. We compare two whitening methods, zero-phase component analysis (ZCA) whitening [27] and robust singular value decomposition (SVD)-based whitening [28]. The latter retains the leading 384 components and discards low-variance directions.

Based on the whitened features, we compute two anomaly scores. The first is the local-density (LD) k NN score [8], which measures

the proximity of a test embedding to the normal reference bank while normalizing the distance according to the local density around its nearest reference sample. The second is relative Mahalanobis distance (RMD) [29], which measures the difference between the distance to the normal mean of the corresponding source or target domain and the distance to the normal mean computed from all source and target normal samples in the whitened feature space. For each scoring method, anomaly scores are computed separately using the source-domain and target-domain reference banks. The source- and target-domain scores are first z -normalized and reduced to a single score by taking their minimum, yielding $\overline{\text{LD}}$ and $\overline{\text{RMD}}$ for LD and RMD, respectively. The resulting scores are then z -normalized again and combined by a weighted sum to obtain the layer-level anomaly score,

$$S_{e,\ell} = w_{\text{LD}} z(\overline{\text{LD}}) + (1 - w_{\text{LD}}) z(\overline{\text{RMD}}), \quad (4)$$

where $z(\cdot)$ denotes z -normalization, $e \in \mathcal{E}$ indexes the encoders, and $\ell \in \mathcal{L}_e$ indexes the selected layers of encoder e . The final clip-level score S is obtained by averaging the layer-level scores within each encoder and then averaging across encoders,

$$S = \frac{1}{|\mathcal{E}|} \sum_{e \in \mathcal{E}} \frac{1}{|\mathcal{L}_e|} \sum_{\ell \in \mathcal{L}_e} S_{e,\ell}. \quad (5)$$

For decision making, we apply the same scoring pipeline to the source-domain normal training clips of each machine and set the decision threshold to the 90th percentile of the resulting scores. A test clip is classified as anomalous when its score S exceeds the threshold of the corresponding machine.

3. EXPERIMENTS

3.1. Experimental Setup

We train the adapters using normal clips from the DCASE 2026 Task 2 development training set [12]–[14]. The resulting adapted encoders are denoted by “+VICReg” and “+ArcFace”, respectively. In addition, the ArcFace adaptation is further continued using the DCASE 2026 additional-training set [12], and the resulting encoder is denoted by “+ArcFace-Additional”. Only the near-field channel is used throughout the experiments. The DCASE 2026 Task 2 development test set is used for in-domain evaluation because its machine types have been observed during adapter training. We additionally evaluate the adapted encoders on the labeled DCASE 2023–2025 Task 2 evaluation sets [2]–[4], [30]–[32]. We refer to this setting as cross-domain evaluation because the cross-year evaluation sets contain machine types and operating conditions that are unseen during adapter training.

All adapters are optimized using AdamW [33], together with a cosine learning-rate schedule and a class-balanced sampler. Following the DCASE Task 2 protocol, performance is evaluated using the official metric Ω^1 .

3.2. Analysis of front-end: encoders, layers, and adaptation

In this part, we use the same back-end for all encoders, consisting of robust SVD-based whitening followed by the fusion of local-density and relative Mahalanobis scores with $w_{\text{LD}} = 0.7$.

Table 2 compares the performance of six pretrained encoders across the evaluated datasets. ATST-Frame, SSLAM, EAT-large, and BEATs perform well on the DCASE 2026 Task 2 development set, with the best Ω reaching 65.87%, but their performance drops to approximately 50% on the DCASE 2025 Task 2 evaluation set. In contrast, CED and AST achieve lower Ω values on the DCASE 2026

¹<https://dcase.community/challenge2026/task-first-shot-unsupervised-anomalous-sound-detection-for-machine-condition-monitoring>

Table 2: Ω (%) of pretrained encoders with different embedding layers. “23~25Avg.” is the average over the DCASE 2023–2025 Task 2 evaluation sets.

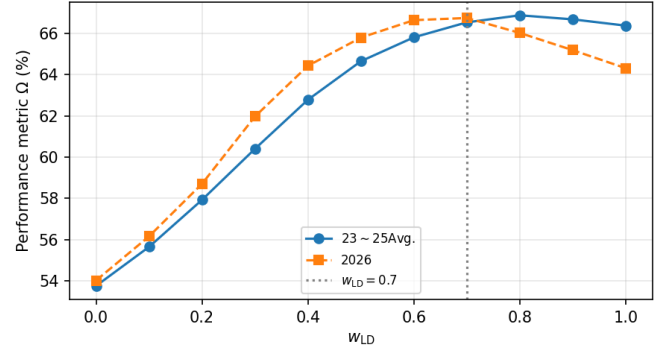
Encoders	Layers	2026	2023	2024	2025	23~25Avg.
ATST-Frame	Final	57.98	61.41	57.14	49.56	56.04
	Best	65.30	70.80	59.51	49.84	60.05
	{0, 1, 2}	65.87	70.85	60.11	49.39	60.11
	All	63.27	68.46	60.72	49.39	59.52
SSLAM	Final	63.60	67.25	62.43	49.07	59.58
	Best	63.60	67.25	62.43	49.07	59.58
	{2, 10, 11}	65.18	70.39	63.43	49.46	61.10
	All	64.03	70.62	63.78	49.70	61.37
EAT-large	Final	60.39	67.05	62.21	49.33	59.53
	Best	62.67	71.72	63.81	49.72	61.75
	{3, 11, 12}	64.29	72.00	64.88	50.21	62.37
	All	62.97	71.15	63.75	49.76	61.55
BEATs	Final	59.28	60.22	58.30	49.95	56.16
	Best	62.77	70.67	61.23	50.61	60.83
	{5, 7, 8}	63.53	70.48	62.83	50.33	61.22
	All	63.39	70.99	62.84	50.34	61.39
CED	Final	57.09	63.26	60.29	61.28	61.61
	Best	60.20	67.61	61.77	60.96	63.45
	{5, 6, 7}	60.70	68.43	62.36	61.13	63.98
	All	60.47	67.52	63.68	61.34	64.18
AST	Final	55.97	61.84	60.67	62.55	61.69
	Best	59.91	67.29	61.15	60.08	62.84
	{2, 3, 5}	59.40	68.28	61.97	60.90	63.72
	All	58.39	66.83	63.05	62.11	64.00

Table 3: Ω (%) of the adapted models using re-selected layers with SSLAM {2, 3, 4}, EAT-large {3, 13, 14} and BEATs {5, 6, 7}. Pretrained results using the best three layers from Table 2 are also included for comparison.

Encoders	Adaptation	2026	2023	2024	2025	23~25Avg.
SSLAM	Pretrained	65.18	70.39	63.43	49.46	61.10
	+VICReg	64.84	74.69	63.92	61.64	66.75
	+ArcFace	64.75	72.81	63.26	61.96	66.01
	+ArcFace-Add.	64.39	73.23	63.76	61.57	66.19
EAT-large	Pretrained	64.29	72.00	64.88	50.21	62.37
	+VICReg	62.96	71.94	63.56	60.78	65.42
	+ArcFace	66.51	74.16	63.76	60.29	66.07
	+ArcFace-Add.	66.52	74.55	63.20	60.73	66.16
BEATs	Pretrained	63.53	70.48	62.83	50.33	61.22
	+VICReg	60.51	70.36	60.58	59.67	63.53
	+ArcFace	63.57	71.54	63.18	61.76	65.49
	+ArcFace-Add.	63.97	71.86	62.83	62.04	65.58

Task 2 development set but maintain similar performance on the DCASE 2025 Task 2 evaluation set. These results indicate that strong in-domain performance does not necessarily translate into better cross-domain generalization. Moreover, the best-performing encoder varies across datasets, which motivates combining multiple encoders to improve robustness across evaluation conditions. Table 2 also shows that aggregating the three selected layers outperforms using only the final layer for all six encoders on the DCASE 2026 Task 2 development set. The preferred representation depth varies across encoders, indicating that a fixed layer-selection strategy may not be optimal for all models.

We further investigate the effect of LoRA-based adaptation on ASD performance, as shown in Table 3. The main benefit of adaptation is observed in cross-domain generalization. All three pretrained encoders show substantial performance degradation on the DCASE 2025 Task 2 evaluation set, whereas LoRA adaptation raises their performance to approximately 60%–62%. Its effect on the DCASE 2026 Task 2

**Fig. 2:** Influence of w_{LD} on performance using the “Full-Ensemble” encoder, where $w_{LD} = 0$ corresponds to RMD and $w_{LD} = 1$ corresponds to LD.

development set is smaller and varies across encoders. Both adaptation objectives improve cross-domain performance, although the gains vary across encoders. VICReg achieves the highest average performance for SSLAM, whereas ArcFace yields more consistent gains for EAT-large and BEATs, particularly on the cross-domain evaluation sets, while maintaining comparable in-domain performance. Furthermore, LoRA adaptation changes the location of the best layers. For example, the selected layers for SSLAM shift from 2, 10, 11 before adaptation to 2, 3, 4 after adaptation, while those for EAT-large shift from 3, 11, 12 to 3, 13, 14. Accordingly, we re-select the layers on the DCASE 2026 Task 2 development set after adaptation rather than inheriting the layer choices of the pretrained encoder.

3.3. Analysis of back-end: whitening, scoring and fusion

In this part, we evaluate twelve encoder variants. For each encoder, embeddings are extracted from its three selected layers. The corresponding results are reported in Table 4.

One can see from the results that removing whitening consistently degrades performance, with a larger effect on the in-domain evaluation. Under the Full-Ensemble setting, removing whitening reduces the LD performance by approximately 5 points and also substantially degrades the RMD performance. Robust-SVD whitening provides a small advantage over ZCA whitening. Among the evaluated individual scoring methods, LD yields higher performance than RMD across the reported conditions, with gains of approximately 8–13 percentage points in 23~25Avg.

Fusion with RMD mainly benefits in-domain performance, while providing limited gains on cross-domain evaluation. For example, under robust-SVD whitening, the performance of the Full-Ensemble increases from 64.3% to 66.7% on the DCASE 2026 Task 2 development set, whereas the gains on the cross-domain evaluation sets remain limited. Figure 2 shows that $w_{LD} \approx 0.7$ achieves the best performance on the DCASE 2026 Task 2 development set.

3.4. Ensembling

Table 5 reports the results of encoder ensembles formed by averaging encoder-level anomaly scores. In this part, we use a fixed back-end consisting of robust SVD-based whitening followed by the fusion of LD and RMD with $w_{LD} = 0.7$. Compared with the best single encoder, the fixed Nine-Ensemble increases Ω from 64.39% to 66.98% on the DCASE 2026 Task 2 development set and from 66.19% to 66.68% on the DCASE 2023–2025 Task 2 average. The Full-Ensemble performs slightly worse than the Nine-Ensemble in different settings, indicating that adding more encoders does not necessarily lead to

Table 4: Ω (%) of different encoders with different backends, where “Full-Ensemble” denotes the ensemble of the 12 encoders listed in the table.

Encoders	Dataset	None Whitening			ZCA Whitening			Robust-SVD Whitening		
		LD	RMD	LD+RMD	LD	RMD	LD+RMD	LD	RMD	LD+RMD
ATST-Frame	2026	57.67	51.19	57.73	64.56	53.01	66.18	64.79	53.04	65.87
	23~25Avg.	59.56	50.17	59.03	59.58	51.84	59.56	60.17	52.38	60.11
SSLAM	2026	57.74	50.53	57.81	62.31	54.42	64.92	62.70	54.42	65.18
	23~25Avg.	59.78	51.30	58.90	60.53	53.33	60.77	61.00	53.24	61.10
EAT-large	2026	57.62	51.19	57.54	62.39	52.66	63.85	63.18	52.55	64.29
	23~25Avg.	59.35	49.74	58.43	62.44	52.64	62.20	62.31	53.07	62.37
BEATs	2026	57.03	52.10	57.85	61.57	52.42	62.64	62.62	52.45	63.53
	23~25Avg.	60.08	50.73	59.21	60.62	52.29	60.54	61.65	52.26	61.22
CED	2026	56.22	52.47	56.94	58.03	49.42	57.66	60.64	50.70	60.70
	23~25Avg.	63.23	50.46	62.02	62.45	51.64	62.31	64.57	51.60	63.98
AST	2026	56.85	51.86	57.02	60.22	51.91	60.89	60.21	49.73	59.40
	23~25Avg.	62.58	51.98	61.97	63.19	53.02	63.37	63.92	52.34	63.72
SSLAM + ArcFace	2026	59.01	50.47	58.77	62.49	53.72	64.04	62.90	55.00	64.75
	23~25Avg.	63.13	50.66	62.29	65.90	53.45	65.81	66.29	53.46	66.01
SSLAM + ArcFace-Add.	2026	59.24	51.52	59.11	63.05	55.12	64.75	62.82	55.05	64.39
	23~25Avg.	63.23	49.75	62.28	66.34	53.41	66.00	66.66	53.38	66.19
EAT-large + ArcFace	2026	61.49	49.99	60.62	64.55	55.96	66.67	65.10	55.12	66.51
	23~25Avg.	62.78	50.66	62.01	66.43	53.86	66.19	66.39	53.93	66.07
EAT-large + ArcFace-Add.	2026	61.04	49.32	60.00	64.32	56.54	66.68	64.12	56.55	66.52
	23~25Avg.	63.79	50.11	62.81	66.45	53.85	66.43	66.30	53.72	66.16
BEATs + ArcFace	2026	58.00	52.18	58.61	61.91	52.47	63.12	62.80	51.46	63.57
	23~25Avg.	64.20	51.05	63.18	64.53	53.10	64.49	66.02	53.58	65.49
BEATs + ArcFace-Add.	2026	57.49	52.05	57.96	62.46	52.32	63.24	63.36	52.66	63.97
	23~25Avg.	64.24	50.95	63.39	64.99	53.26	64.85	66.04	53.76	65.58
Full-Ensemble	2026	59.30	51.91	59.85	63.57	54.54	66.23	64.26	54.01	66.74
	23~25Avg.	64.98	50.85	63.96	65.43	53.65	65.96	66.31	53.71	66.53

Table 5: Ensemble Ω (%). The “-family” refers to a pretrained encoder and its two adapted variants. “Nine-Ensemble” combines three encoder families. “Greedy-Ensemble” is constructed through greedy encoder selection.

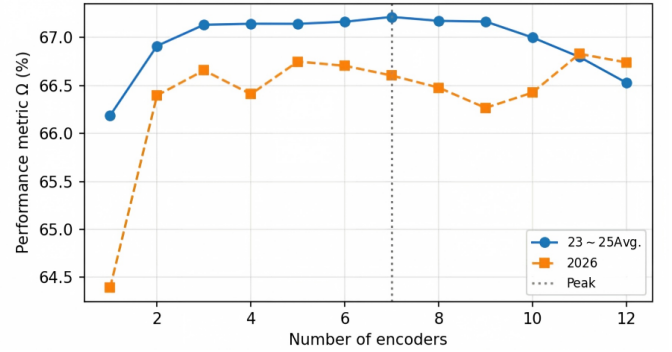
Encoders	2026	2023	2024	2025	23~25Avg.
Best single encoder	64.39	73.23	63.76	61.57	66.19
SSLAM-family	65.89	73.24	63.97	60.62	65.94
EAT-large-family	66.73	74.76	64.62	59.82	66.40
BEATs-family	64.02	71.66	63.08	60.92	65.22
Nine-Ensemble	66.98	74.89	64.51	60.65	66.68
Full-Ensemble	66.74	74.18	64.62	60.78	66.53
Greedy-Ensemble	66.60	74.81	64.78	62.05	67.21

better performance. The within-family ensembles show that the effect of ensembling varies across encoder backbones.

Figure 3 further analyzes the effect of encoder selection on ensemble performance using greedy additive ensembling. Specifically, at each step, we add the remaining encoder that yields the highest average Ω on the DCASE 2023–2025 Task 2 evaluation sets among all candidate additions when combined with the currently selected encoders. The results show that the greedy additions initially improve ensemble performance, whereas further additions eventually degrade performance. On the DCASE 2023–2025 Task 2 evaluation sets, the best Ω achieved by this strategy exceeds that of the Full-Ensemble, indicating that adding more encoders does not necessarily improve ensemble performance. However, as shown in Table 5, the Greedy-Ensemble does not retain the same advantage on the DCASE 2026 Task 2 development set, indicating that the benefit of this selection does not generalize consistently across evaluation conditions.

4. CONCLUSION

We conducted an empirical study of the key design choices in first-shot ASD. Our results show that strong in-domain performance

**Fig. 3:** Greedy additive ensembling of the encoders in Table 4. At each step, the remaining encoder giving the highest average Ω on the DCASE 2023–2025 Task 2 evaluation sets after addition to the current ensemble is selected.

does not necessarily imply better cross-domain generalization. For representation learning, aggregating selected layers is more effective than relying only on the final layer on the DCASE 2026 Task 2 development set, while LoRA-based adaptation mainly improves cross-domain performance. Encoder ensembling can provide additional gains, but its effectiveness depends on which encoders are combined rather than simply on ensemble size. For anomaly scoring, whitening substantially improves performance. LD performs better than RMD across the evaluated conditions, while RMD can provide additional benefit when fused with LD for in-domain evaluation. However, the in-domain evaluation has some limitations. The layer selection and fusion weight were determined using the DCASE 2026 Task 2 development set, so the corresponding in-domain results may be affected by tuning bias. Nevertheless, the cross-domain evaluation on previous DCASE datasets provides additional evidence for the robustness of the findings.

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